

Abstract

Rational design of single-atom catalysts is essential for achieving efficient and sustainable energy conversion. In particular, precise atomic-level control over active sites and defect structures is crucial for electrocatalytic systems operating in acidic media. Moreover, well-defined surface architectures enable deeper mechanistic insights into catalytic processes. Herein, we report a single-atomic-site platinum catalyst stabilized on defect-engineered $\text{Mo}_2\text{TiC}_2\text{T}_x$ MXene (denoted as $\text{Pt@V}_{\text{Mo}}/\text{MXene}$), where Pt atoms are selectively anchored at molybdenum vacancy sites. By integrating electrochemical potential control with precise regulation of defect formation, the local coordination environment is modulated to effectively stabilize isolated Pt atoms at catalytically active sites.

I. One-step potential control with increasing HER performance

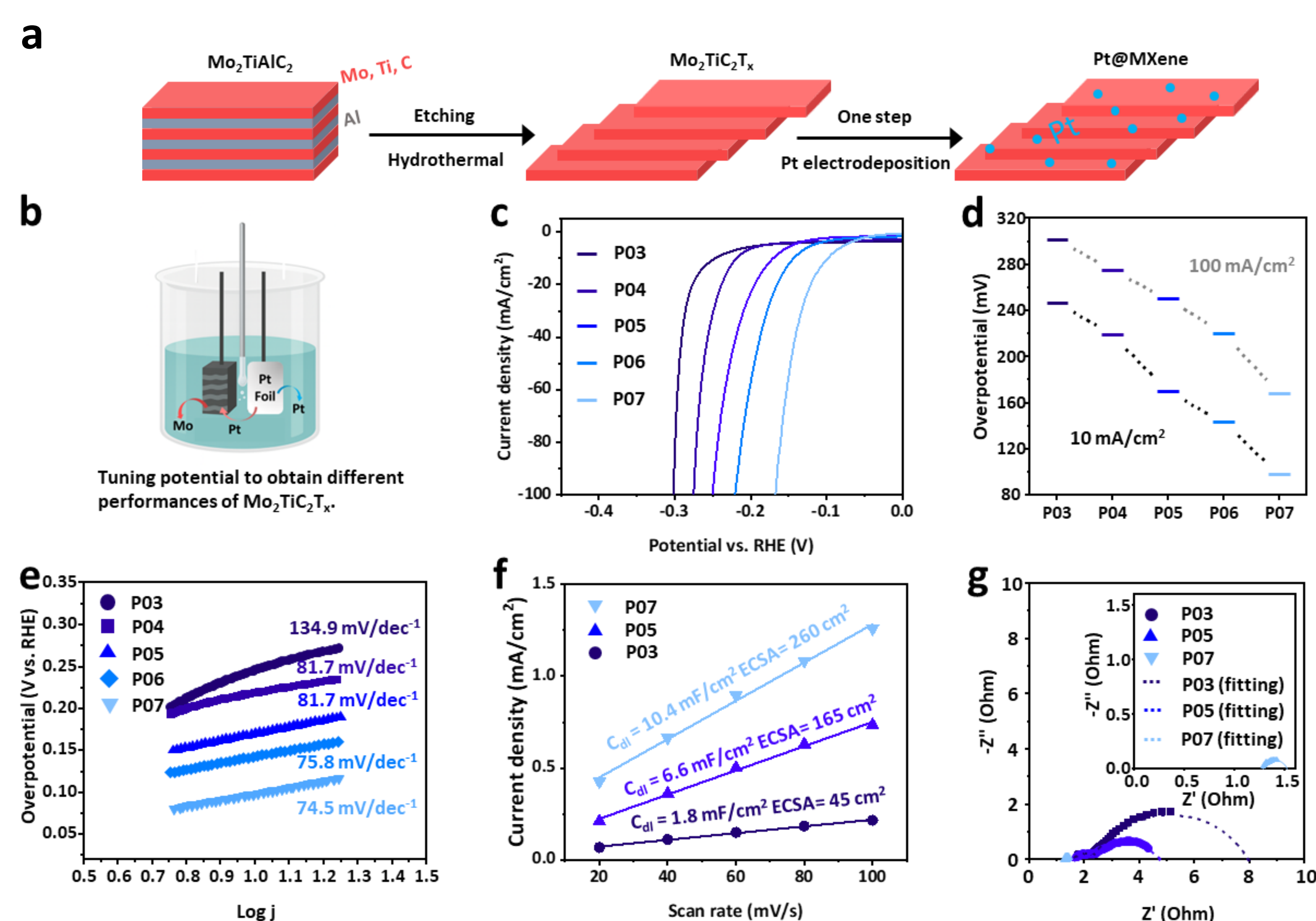


Figure 1. (a, b) Schematic of preparation for one-step Pt@MXene . (c) HER LSV curves of different potential treated MXene. (d) Overpotential at 10 and 100 mA/cm^2 . (e) Tafel plots, (f) ECSA, and (g) Nyquist plots of different potential treated MXene. (Samples are labeled as POX , where X denotes the end potential in cyclic voltammetry (CV). All CV scans start from 0 V vs. RHE, using a Pt plate as the counter electrode. For example, P07 represents a CV range of 0 to -0.7 V vs. RHE.)

II. Characterization of potential-driven Pt migration

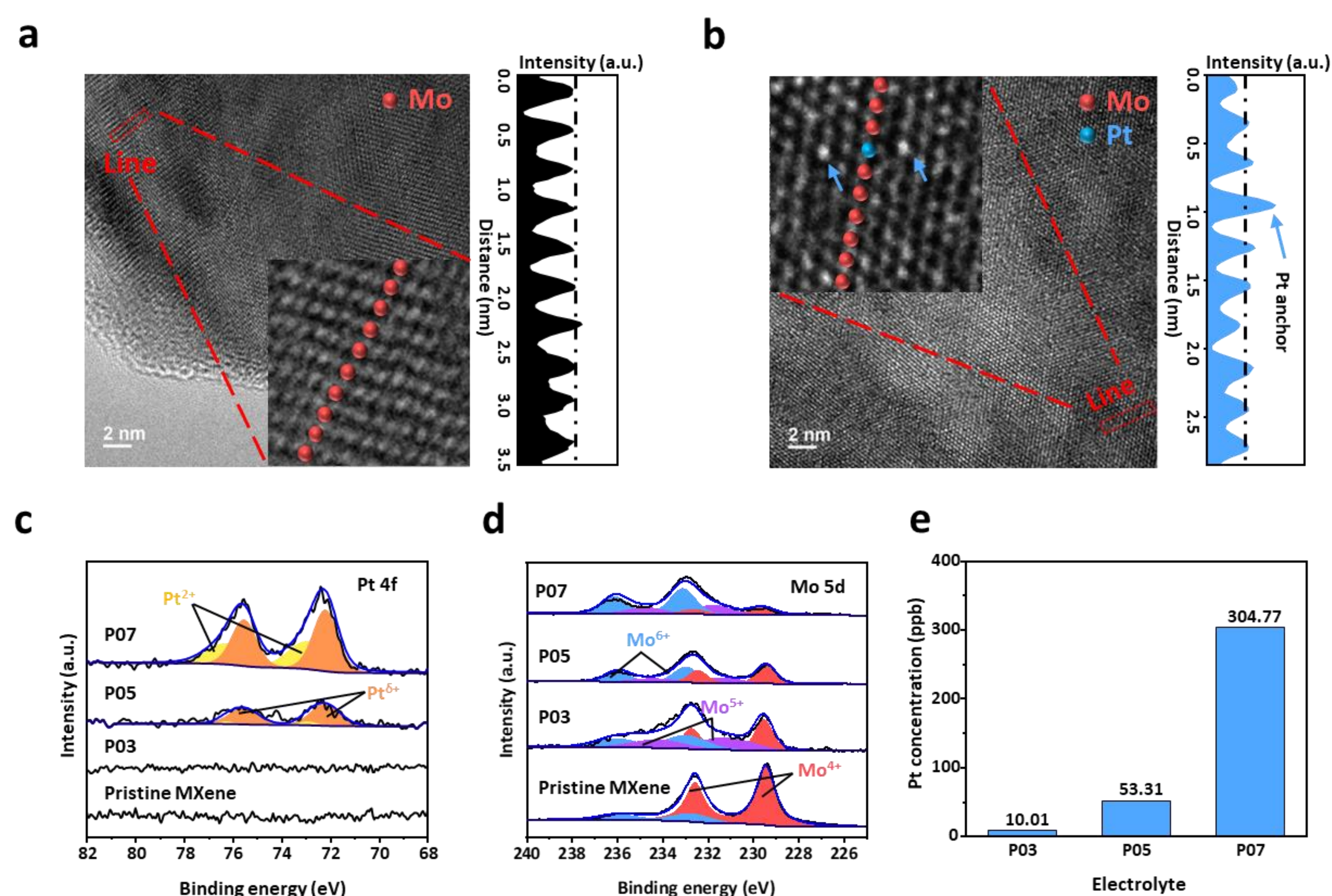
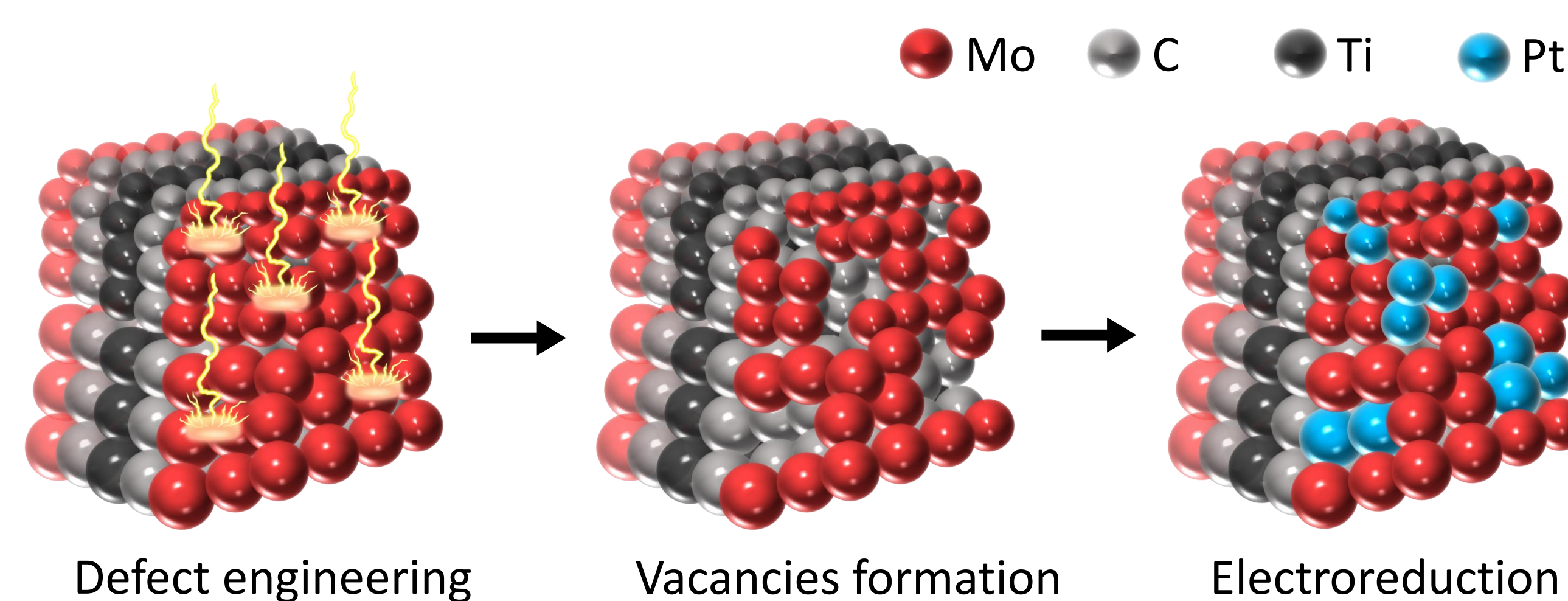


Figure 2. HR-TEM image and intensity profile of (a) Pristine MXene and (b) High potential treated Pt/MXene . XPS spectra of (c) Pt 4f and (d) Mo 5d under increasing potentials. (e) Pt concentration in electrolyte measured by ICP.



III. Two-step strategy : Potential-induced Mo vacancy formation

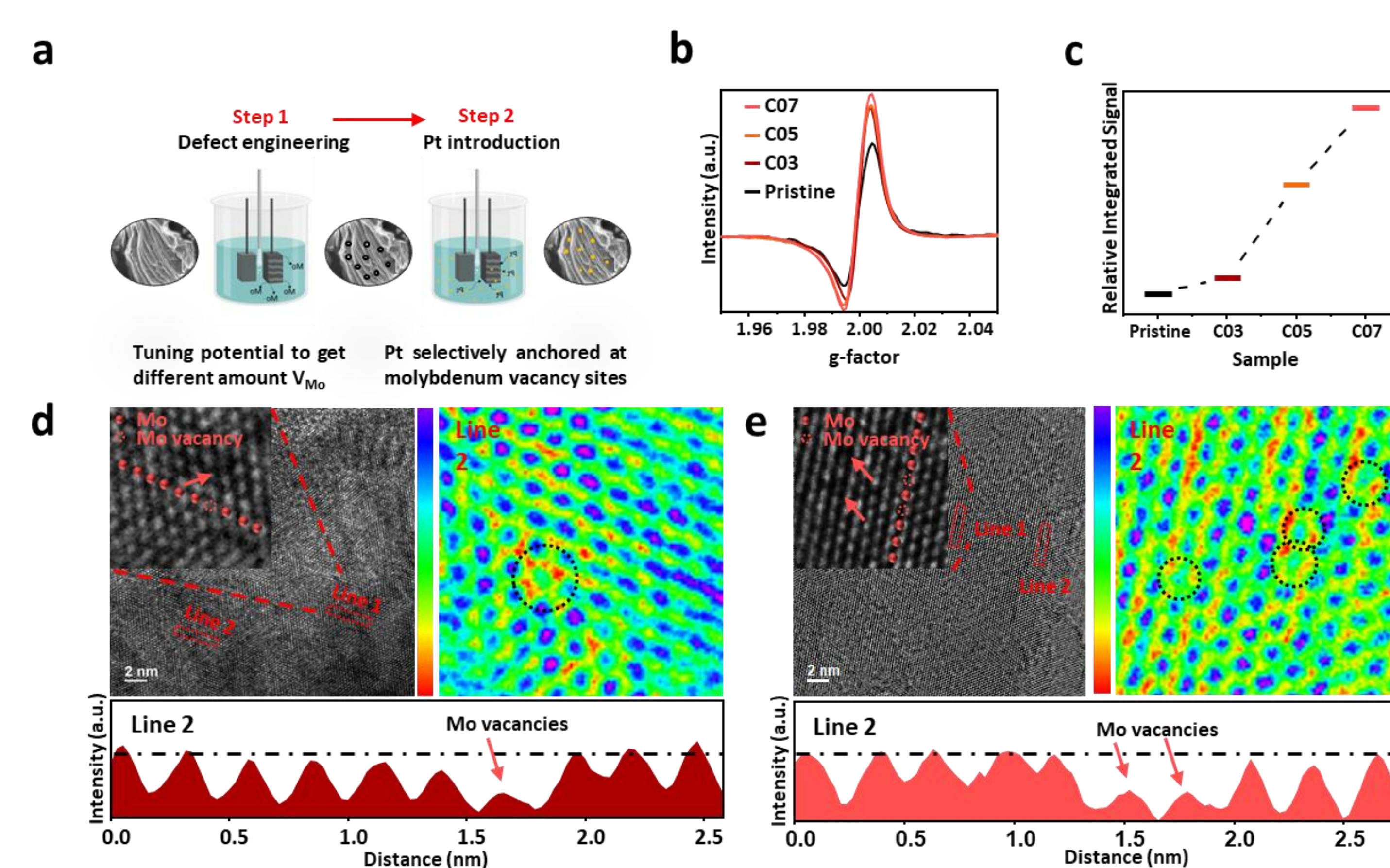


Figure 3. (a) Schematic of preparation for two-step $\text{Pt@V}_{\text{Mo}}/\text{MXene}$. (b) EPR detection of vacancy concentration and (c) relative integrated signal. HR-TEM image, false-colored HRTEM image, and intensity profile of (d) Low potential and (e) High potential treated $\text{V}_{\text{Mo}}/\text{MXene}$. (Samples are labeled as COX , where X denotes the end potential in cyclic voltammetry (CV). All CV scans start from 0 V vs. RHE, using a Pt plate as the counter electrode.)

IV. Two-step strategy : Pt selectively anchored

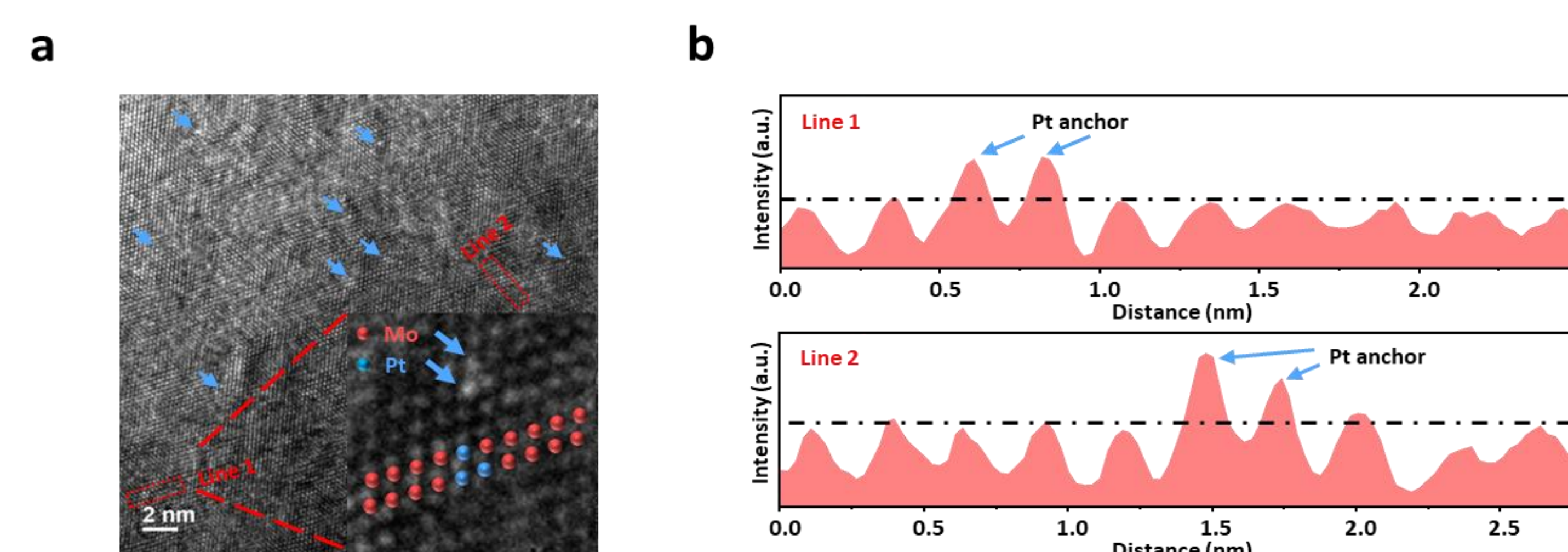


Figure 4. (a) HR-TEM image and (b) intensity profile of $\text{Pt@rich-V}_{\text{Mo}}/\text{MXene}$.

V. Advantages of generate rich vacancies before Pt anchor

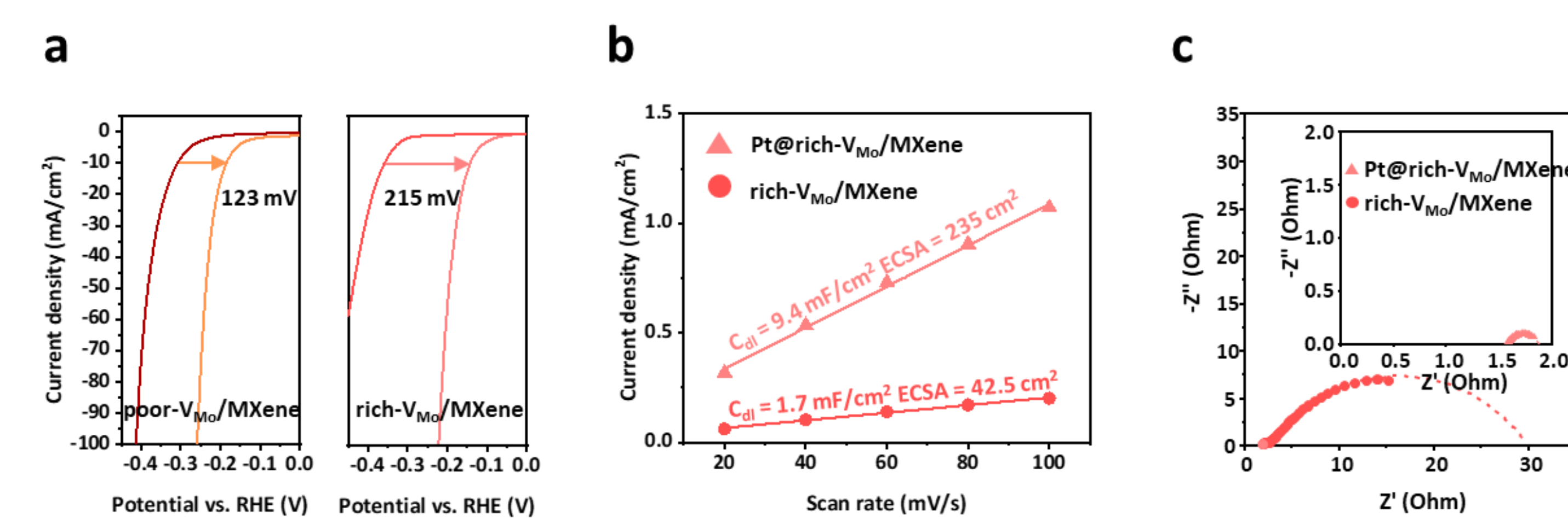


Figure 5. (a) LSV curves of HER performance increase before and after Pt selectively anchored decided by different vacancy concentration. (b) ECSA and (c) Nyquist plots of $\text{rich-V}_{\text{Mo}}/\text{MXene}$ and $\text{Pt@rich-V}_{\text{Mo}}/\text{MXene}$.

Conclusion

This work demonstrates that electrochemical potential serves as a precise driving force to modulate Pt anchoring on MXene, significantly improving HER activity. Through a controlled defect-engineering strategy, Mo vacancies are generated in a tunable manner, directly governing Pt migration and stabilization. A stepwise potential design eliminates ambiguity from potential-induced Pt ion fluctuations, enabling a clear mechanistic correlation between vacancy density and Pt loading. Consequently, HER performance increases from $\sim 40\%$ to $\sim 60\%$, revealing a strong vacancy–metal synergistic effect.

References

- [1] Zhang, J., Zhao, Y., Guo, X. et al. Single Platinum Atoms Immobilized on an MXene as an Efficient Catalyst for the Hydrogen Evolution Reaction. *Nat. Catal.* **2018**, 1, 985–992.